

Soru 1) Komut Seti Mimarisi

a)

	A=62		B=-75	
İkili gösterim	0011 1110	V (taşma)	1011 0101	V (taşma)
Aritmetik Sola Öteleme	0111 1100	0	0110 1010	1
Aritmetik Sağa Öteleme	0001 1111	0	1101 1010	0

b)

Mnemonic	Durum	Test	A= 1101 1110 B= 0101 1110	A= 0101 0010 B= 1101 0001
(A-B) sonucu			1000 0000	1000 0001
N, Z, V			N=1, Z=0, V=0	N=1, Z=0, V=1
BG	$A > B$	$(N \oplus V) + Z = 0$	False	True
BGE	$A \geq B$	$N \oplus V = 0$	False	True
BL	$A < B$	$N \oplus V = 1$	True	False
BLE ¹	$A \leq B$	$(N \oplus V) + Z = 1$	True	False

¹ BLE satırı sınavda $(N \oplus V) + Z = 0$ olarak verildiği için iptal.

Soru 2)

	DA	AA	BA	MB	FS	MD	RW
$R7 \leftarrow 0$	111	000	000	0	1010	0	1
$R5 \leftarrow sl\ R7$	101	---	111	0	1110	0	1
$R5 \leftarrow \text{Data in}$	101	---	---	-	---	1	1
$R4 \leftarrow sr\ R4$	100	---	100	0	1101	0	1
$R3 \leftarrow R1 - \text{constant in}$	011	001	---	1	0101	0	1
$R2 \leftarrow R2 + 1$	010	010	---	-	0001	0	1
$R1 \leftarrow R2 \oplus R3$	001	010	011	0	1010	0	1
$R3 \leftarrow R4 + R5$	011	100	101	0	0010	0	1

Encoding of Control Word for the Datapath									
DA, AA, BA		MB		FS		MD		RW	
Function	Code	Function	Code	Function	Code	Function	Code	Function	Code
$R0$	000	Register 0	0	$F = A$	0000	Function 0	0	No Write 0	0
$R1$	001	Constant 1	1	$F = A + 1$	0001	Data in 1	1	Write 1	1
$R2$	010			$F = A + B$	0010				
$R3$	011			$F = A + B + 1$	0011				
$R4$	100			$F = A + \bar{B}$	0100				
$R5$	101			$F = A + \bar{B} + 1$	0101				
$R6$	110			$F = A - 1$	0110				
$R7$	111			$F = A$	0111				
				$F = A \wedge B$	1000				
				$F = A \vee B$	1001				
				$F = A \oplus B$	1010				
				$F = \bar{A}$	1011				
				$F = B$	1100				
				$F = sr\ B$	1101				
				$F = sl\ B$	1110				